



Energy Communities in Ports and City-Port Hinterlands: A New Paradigm for a Green Harbor

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Abstract: Port cities and their hinterlands present particular challenges for sustainable development, including high energy demand, urban heat island (UHI) effects, energy poverty, and environmental impacts from port operations. This work aims at verifying two hypotheses: first, that increasing renewable energy community (REC)/self-consumption scheme (SCS) membership significantly improves self-consumption performance for a fixed photovoltaic (PV) installation; and second, that the spatial distribution of impervious surfaces in port hinterland areas is directly correlated with UHI intensity and therefore highlights priority zones for Nature-based Solution (NbS) intervention. An hourly photovoltaic simulation model is used to evaluate the effect of community size on the self-consumption factor, and a geographic information system analysis combining Copernicus reanalysis temperature data and high-resolution land cover layers is used to characterize UHI dynamics in Genoa. The implementation framework is provided by the EnerCmed project, which is co-funded by the European Union (EU)’s Interreg Euro-MED programme. The simulation results demonstrate that with an increase in REC/SCS membership from 1 to 30 apartments, the self-consumption factor increases from 5% to 50% for a 50 kWp installation. The results of the UHI analysis show nocturnal summer intensification of +3 to +4 °C in the Genoa port hinterland area, with strong correlation with impervious surface density. The results show how renewable energy governance coupled with nature-based climate adaptation strategies can simultaneously improve renewable energy self-consumption and mitigate urban heat stress in port cities’ hinterlands. These findings support the new hybrid approach as an effective replicability strategy for green harbors and sustainable Mediterranean port cities.

Keywords: Renewable energy community; Nature-based Solutions; Urban heat island; Port city hinterland; Mediterranean; Green harbor; Energy poverty; Self-consumption

1 Introduction

Ports are vital hubs in global trade but pose significant environmental challenges, including high energy consumption and emissions [1]. Energy efficiency has emerged as a hot research topic in addressing sustainable development goals in the maritime industry and port sector [2, 3]. City-port hinterlands, that support port operations, have their own energy demands related to the diverse types of buildings in the area (including civil habitations, public buildings, schools, hospitals, and commercial buildings). Addressing these energy demands with a sustainability perspective is crucial for reducing the environmental footprint of port cities [4].

Energy communities are collaborative arrangements where local stakeholders generate, share, and manage energy. These communities can enhance energy independence, reduce costs, and lower emissions, with key features including

local renewable energy generation, shared energy resources, and active community engagement [5]. A recent study mapping the Renewable Energy Communities in Europe shows its increasing deployment and diffusion [6]. Energy communities represent a transformative approach to managing energy consumption and production, particularly in strategic areas like ports and their hinterlands; a review of recent trends, system modeling, business models, and optimization objectives can be found in [7]. These communities offer valuable benefits in the efforts to achieve the sustainability goals set by the European Union (EU) [8]. Renewable energy communities (REC) and other energy sharing configurations, such as self-consumption schemes (SCS), utilize local renewable energy resources, optimize energy use, and engage local stakeholders, contributing to both environmental sustainability and economic resilience [9]. Implementing the REC/SCS concept could be part of the roadmap for future green harbors in Mediterranean ports, leveraging the region's renewable energy potential and increasing the interest of stakeholders at various levels, from local to EU.

This study will explore two particular ideas behind the hybrid REC/SCS + Nature-based Solutions (NbS) paradigm. The first is that increasing the number of members that join a REC, producing renewable energy with a photovoltaic (PV) plant, will have a significant positive effect on the performance of the self-consumption system. The second is that the distribution of impervious surfaces in hinterland areas of ports is directly related to the intensity of the urban heat island (UHI) effect, and thereby a spatial study of the port-city hinterlands is useful for identifying areas where NbS could be applied. The potential of the REC/SCS + NbS hybrid approach will be presented within the transnational framework of the European project EnerCmed, which has been applied across five Mediterranean cities. The novelty of the present approach with respect to existing literature on REC/SCS applications in urban contexts is that it provides not only a theoretical framework but also insights for real deployments of pilot actions, especially in port cities' hinterland areas. The study seeks to provide a comprehensive roadmap for establishing green harbors that contribute to the EU's sustainability goals and serve as models for sustainable urban development.

2 Methodology

This study adopts a multi-component approach to investigate the feasibility and benefits of implementing the hybrid REC/SCS + NbS paradigm in Mediterranean port-city hinterlands. To assess the paradigm, three complementary methods are employed:

- An energy simulation model for REC/SCS performance assessment,
- A geospatial analysis of the UHI effect,
- A transnational implementation framework, tested across five pilot cities through the EnerCmed project.

2.1 Renewable Energy Community/Self-Consumption Scheme Simulation and Energy Balance Model

Ports offer substantial potential for renewable energy installations, such as solar panels in warehouses or wind turbines along coastlines. In addition, energy communities in ports can optimize energy use, reduce operational costs, and mitigate environmental impacts.

To estimate the benefits coming from sharing PV electricity in an REC/SCS, an energy balance model is proposed. The model is used to calculate the shared energy, comparing the produced energy (hour by hour) by a PV plant, and the consumption of the REC/SCS members. To estimate the REC/SCS performance, it is possible to calculate the dimensionless self-consumption factor, a , that represents the efficient use of the produced energy by the PV plant, as in Eq. (1). In the equation is necessary to consider the estimation of produced power during a typical year (E_p in kWh) and the consumption of the cluster of apartments that join the SCS as members, composed of real and virtual self-consumption ($E_{sc \text{ real}}$ and $E_{sc \text{ virtual}}$, respectively, both in kWh).

$$a = \frac{E_{sc}}{E_p} = \frac{E_{sc \text{ real}} + E_{sc \text{ virtual}}}{E_p} \quad (1)$$

Considering, as an example, a residential complex forming a REC/SCS located in Genoa the energy balance model is applied under different scenarios. The model simulates hourly PV energy production and residential consumption profiles over a full reference year for a rooftop installation in the selected location. Solar irradiance data were obtained from the free application Photovoltaic Geographical Information System (PVGIS) [10] and residential consumption profiles from the Autorità di Regolazione per Energia Reti e Ambiente (ARERA) [11]. In this study, three community size configurations were simulated for a reference PV plant of 50 kWp: a single apartment, a REC of 10 apartments, and a REC of 30 apartments.

2.2 Nature-Based Solutions and Urban Heat Island Effect

The UHI effect was studied by comparing the temperature maps and the surface cover of Genoa using GIS tools. The images were obtained and adapted from the Copernicus EU mission, the UHI temperature map [12], and the land use, combining the high-resolution imperviousness and tree cover layers, using 2018 satellite data [13]. Genoa was

used as a case study area because of its representative Mediterranean port-city morphology, high imperviousness in the coastal strip adjacent to the city, and the presence of one EnerCmed pilot intervention site.

2.3 Transnational Approach to Implementing the Hybrid Paradigm

The context for the transnational implementation of the study is provided by the EnerCmed project, co-financed by the EU within the Interreg Euro-MED programme. The project focuses on the design of pilot actions in the hinterland of Mediterranean ports across five port-cities (Valencia, Patras, Pula, Novigrad, and Genoa), integrating the installation of PV plants within REC/SCS schemes with the implementation of an NbS, such as green roofs or gardens. The pilot actions serve to test the REC/SCS + NbS hybrid paradigm across different national frameworks and climates.

3 Results

This section presents the results of the three analytical components of the study: the REC self-consumption scaling analysis, the UHI characterization of the Genoa port-city case study, and the implementation status and expected outcomes of the EnerCmed pilot actions across five Mediterranean port cities.

3.1 Renewable Energy Community/Self-Consumption Scheme Self-Consumption Performance: Scaling Analysis

The simulation results demonstrate the effect of community size on self-consumption performance, as summarized in Table 1. In the baseline scenario, with no community (single user), the self-consumption factor a is approximately $a \approx 0.05$; only 5% of the solar energy produced is consumed on-site, with the remainder exported to the grid. This result is consistent with the broader literature on individual PV self-consumption, where single-user rates in residential settings drop significantly when the system is oversized relative to individual demand [14]. With a community of 10 members, the factor rises to approximately $a \approx 0.20$, while with 30 members the factor rises to approximately $a \approx 0.50$, sharing in all the 3 cases the same 50 kWp of PV capacity.

Table 1. Self-consumption factor, a , as a function of renewable energy community (REC)/ self-consumption scheme (SCS) community size for a 50 kWp photovoltaic (PV) plant in Genoa

Configuration	Apartments	PV Capacity (kWp)	Self-Consumption Factor, a (-)
No REC (single user)	1	50	≈ 0.05
Small REC	10	50	≈ 0.20
Medium REC	30	50	≈ 0.50

Figure 1 illustrates the hourly energy balance for the three simulated configurations over a full reference year. In all three panels, the blue profile represents the hourly PV production, which follows the characteristic seasonal bell curve of Mediterranean solar irradiance. In the single-apartment scenario (Figure 1a), the consumption profile (in orange) is almost invisible against the scale of production, confirming that a single residential user can absorb only a negligible fraction of the energy generated. As the community size increases to 10 apartments (Figure 1b), the aggregate consumption grows, and the shared energy band (yellow) becomes clearly visible. With 30 apartments (Figure 1c), the shared energy band expands substantially and covers a significant portion of the production profile throughout the year, visually demonstrating the scaling effect quantified in Table 1.

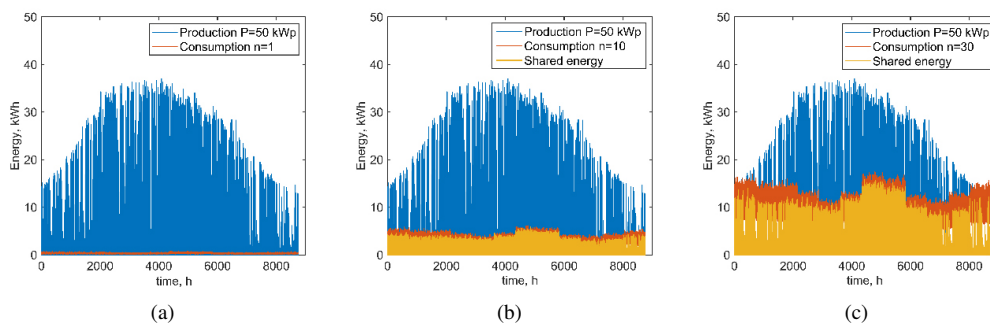


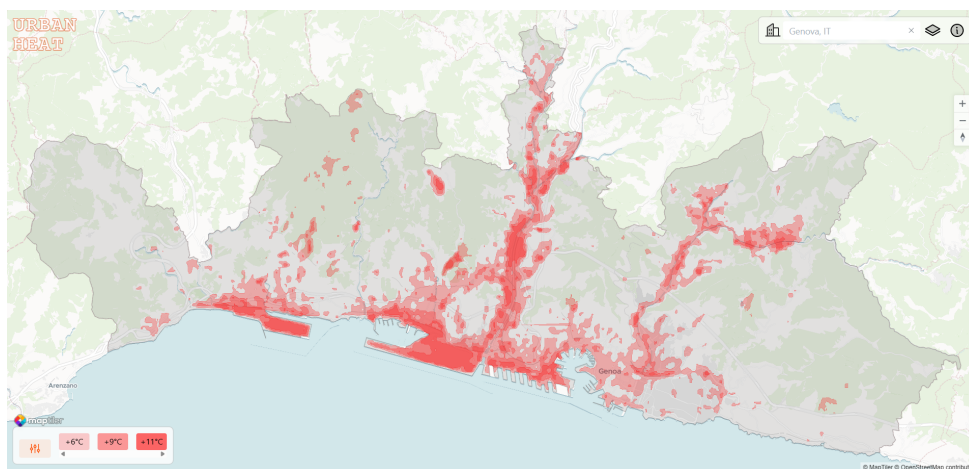
Figure 1. Hourly electrical energy profiles over a full reference year for a 50 kWp photovoltaic plant in Genoa, showing three renewable energy community (REC) configurations: (a) no REC, single apartment ($a \approx 0.05$); (b) REC with 10 apartments ($a \approx 0.20$); and (c) REC with 30 apartments ($a \approx 0.50$)

3.2 Urban Heat Island Characterization: The Genoa Case Study

A major challenge in highly dense urban areas is the UHI effect, a significant increase in temperature that leads to higher energy consumption during summer. In Figure 2, the Genoa case is used to illustrate the correlation between the UHI effect measured during summer (2008–2017) and the surface coverage typology.

During summer days, UHI intensity in urban areas relative to surrounding rural zones tends to remain generally moderate, with values usually below +2 °C, see Figure 2a. During summer nights, UHI intensity significantly increases, with values that can reach +3 to +4 °C in the coastal urban area characterized by a high density of impervious surfaces, corresponding to the port-city's hinterland. This phenomenon has been widely documented in Mediterranean cities, where heat accumulated on urban surfaces during summer days is released at night. The spatial distribution of impervious surfaces in Genoa, as shown in Figure 2b, indicates a direct correlation between impervious surface cover and the intensity of the UHI effect. Intervening on land cover, therefore, means reducing heat accumulation and improving the urban microclimate.

In this context, NbSs represent a strategic tool for intervention, as they directly address the mechanisms that generate the UHI effect by altering the morphology and permeability of spaces and promoting evapotranspiration, therefore reducing energy demand, improving urban comfort, and strengthening the resilience of port cities.



(a)



(b)

Figure 2. (a) Urban heat island (UHI) map for Genoa city, annual maximum land surface temperature; Adapted from [15]; (b) Land classification map of Genoa; Dark red: high imperviousness (port infrastructure, dense urban fabric); Orange: medium-high imperviousness (residential areas); Green: vegetated surfaces; Adapted from [12]

Note: The urban boundary data are provided by Eurostat GISCO, and the temperature data are sourced from NASA Landsat, as visualized in the Urban Heat Island Explorer [15].

3.3 EnerCmed Pilot Implementation and Expected Outcomes

The EnerCmed project, co-funded by the EU under the Interreg Euro-MED programme, provides the transnational framework within which the hybrid paradigm is implemented and tested at scale [16, 17]. The project involves nine partners from four EU member states: Italy, Spain, Greece, Cyprus, and Croatia. The five pilot cities are Genoa, Valencia, Patras, Pula, and Novigrad. They are distributed across the northern Mediterranean coast, as depicted in Figure 3. A sixth city, Larnaca in Cyprus, is also part of this project as a Meta-Replicator, facilitating the transfer and replication of the solutions to other Mediterranean territories.

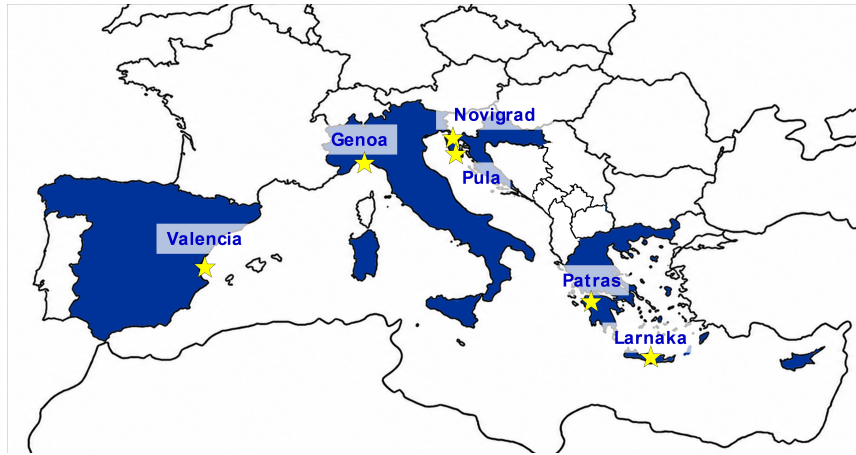


Figure 3. Geographic distribution of the five EnerCmed pilot cities and Larnaca as a Meta-Replicator city across the Mediterranean region; Blue indicates the European countries involved into EnerCmed project

The pilot actions include activating six REC or SCS coupled with NbS in marginalized port neighborhoods with a risk of energy poverty. These cities cover four national regulatory environments. Additionally, they offer a variety of port city typologies, from industrial cities to smaller coastal towns. The pilot cities were selected based on renewable energy potential, port-city strategic importance, and evidence of local public stakeholder involvement. This enables a comparative evaluation approach across a variety of national regulatory, climatic, and social contexts [17].

The project is structured around three work packages (WP) as summarized in Table 2. WP1 focuses on building a methodological and normative framework. This includes the creation of a Knowledge Facility Instrument (KFI), a structured approach to knowledge management, intending to provide scientific and technical support to REC/SCS and NbS implementation [18]. WP2 covers the implementation phase in the five pilot cities, while WP3 covers replication & scaling-up with a multi-city program, with Larnaca (Cyprus) as a Meta Replicator city.

Table 2. EnerCmed project work package (WP) structure

Work Package	Title	Main Activities
WP1	Methodological framework	Development of the transnational renewable energy community (REC)+ Nature-based Solution (NbS) paradigm; Knowledge Facility Instrument (KFI); Terms of Reference (ToR) for REC activation; Policy Brief; shared Knowledge Sharing Platform
WP2	Operational experimentation	Installation of photovoltaic plant on public buildings; formal establishment of RECs and self-consumption schemes (SCS); user engagement; definition of legal and economic-management models; integration of NbS for urban heat island (UHI) mitigation and air quality improvement across five pilot cities
WP3	Replication and scalability	Territorial action plans; capacity building programme through Virtual Academy and webinar series; coaching of technical partners; feasibility assessment tools; replication coordinated by Larnaca (Cyprus) as Meta-Replicator city targeting at least 15 follower cities

The transnational governance framework developed through WP1, which includes the KFI [18], the Terms of Reference (ToR) for REC activation, and the shared Knowledge Platform addresses the principal barriers to REC deployment identified in the literature: regulatory complexity across four national frameworks, financial constraints for lower-income households, low stakeholder awareness, and fragmented building ownership in port hinterland neighborhoods [19].

The PV systems to be installed in the five pilot cities have a combined capacity of 243 kWp. All of these systems are to be installed in public buildings, mainly schools, and to engage public entities and ensure maximum

community benefits. As of March 2026, the project has reached its stage of operational implementation. The PV plant installations have been almost completed and tested at the pilot sites notably in Patras and Genoa, with procedures for grid connections currently underway. The NbS interventions are at different stages of design and implementation. Figure 4 shows examples of PV installation work at two pilot sites.



Figure 4. (a) Photovoltaic (PV) installation works at EnerCmed pilot site: Ferrero primary school, Cornigliano district, Genoa, March 2026; Source: Comune di Genoa; (b) Photovoltaic (PV) installation at EnerCmed pilot site: 6th EPAL School Complex, Patras, March 2026; Source: Municipality of Patras

The EnerCmed project implements the hybrid REC/SCS + NbS concept in five Mediterranean port cities, thereby creating a multi-contextual setting to assess its scalability and replicability potential [16, 17]. The project's replication dimension has already exceeded its targets. The Follower Cities Network (WP3) has reached 16 cities across the Mediterranean as of March 2026, demonstrating strong institutional demand for the REC/SCS + NbS model beyond the five pilot sites.

The project's quantitative objectives demonstrate the scale of impact achievable through this transnational approach. The PV plant capacities installed across all five cities will provide an annual renewable electricity output of near 278 MWh, equivalent to approximately 160 tCO₂ in greenhouse gas emissions. The number of households/users who will be directly impacted or covered by the REC/SCS + NbS measures is estimated at 345, with the direct beneficiaries of the REC/SCS measures expected to achieve a reduction in their energy bills of 15–20% [16, 17].

4 Discussion

The results of this study address the two ideas stated before, regarding sharing electricity by a REC/SCS scheme and implementing NbS to mitigate UHI effect, and situate the findings within the broader context of Mediterranean urban sustainability.

4.1 The Scaling Effect of Community Size and Its Implications

The first hypothesis, that an increase in community size will significantly improve self-consumption efficiency with respect to the installed PV plant, was supported by simulation (Table 1). The efficiency of self-consumption, measured by the self-consumption factor α , rises from around 0.05 for a single apartment building to around 0.50 when 30 buildings share a 50 kWp system, a tenfold increase. These results are in line with recent findings on REC/SCS optimization, which indicate that diversification of the demand profile is a key tool for increasing the level of common self-consumption [20]. This research adds a new perspective to earlier findings by showing that the scaling effect can be useful from the perspective of achieving sustainability on city-port hinterlands [19]; however, the scaling effect must be further studied not only from the energy metrics but also considering the associated economic benefits perceived by the community or individual members.

4.2 Urban Heat Island, Nature-based Solution, and the Energy-Climate Nexus in Port Hinterlands

The second hypothesis, that the spatial distribution of impervious surfaces in port hinterland areas is directly correlated with UHI intensity and therefore highlights priority zones for NbS intervention, is confirmed through geospatial analysis of Genoa. The intensity of nocturnal UHI effects, ranging from +3 to +4 °C, recorded in the densely built coastal strip, shows a strong correlation with high imperviousness, as was demonstrated for urban climate conditions in the Mediterranean area [21–23]. This result extends previous work on urban radiation environment [23] by directly linking surface cover characteristics and urban morphology to thermal vulnerability at the neighborhood scale, showing the different relevant conditions to NbS intervention planning.

4.3 Transnational Replication and the Governance Challenge

The geographical distribution of the pilot cities within the four selected national regulatory regimes, i.e., those of Italy, Spain, Greece, and Croatia, reflects the regulatory diversity that is a defining feature of REC/SCS deployment in EU member states [9, 24]. In contrast to the way such diversity is often viewed as an obstacle, EnerCmed addresses it through the KFI [18], which provides a structured and adaptable framework for REC/SCS activation across different national contexts. This governance contribution extends beyond the project itself and offers a replicable tool for future community energy initiatives in the Mediterranean region.

5 Conclusions

This study has investigated the feasibility and benefits of implementing REC/SCS integrated with NbS in the hinterlands of Mediterranean port-cities, using Genoa as a primary case study and the EnerCmed transnational project as the operational framework. The following conclusions can be drawn:

- The self-consumption scaling analysis demonstrates that community size is the decisive parameter in REC/SCS performance. Increasing membership from a single user to 30 members raises the self-consumption factor from approximately 5% to 50% for a reference 50 kWp PV plant, confirming that the community energy model transforms the economics of rooftop PV plants in urban hinterland contexts where individual investment capacity is structurally limited. The anchoring of REC/SCS on public buildings, particularly schools, emerges as a replicable design strategy that simultaneously maximizes energy sharing, ensures public governance, and targets benefits to the most vulnerable segments of port hinterland communities.
- The UHI characterization of Genoa reveals nocturnal summer heat intensities of +3 to +4 °C in the port hinterland zone, driven by the high imperviousness and low tree cover density that characterize the dense urban fabric adjacent to the port. This thermal condition directly amplifies cooling energy demand and the risk of energy poverty among lower-income residents, creating a feedback loop that passive technological interventions alone cannot break. Green roofs, confirmed as thermally effective and financially viable in the Mediterranean context provide a targeted and co-beneficial NbS response, reducing the urban heat load while simultaneously delivering thermal insulation, stormwater management, noise reduction, and biodiversity benefits.
- The combination of REC/SCS and NbS on public building rooftops constitutes a hybrid paradigm that addresses the port hinterland sustainability challenge in an integrated manner, linking energy transition, climate resilience, and social inclusion within a single intervention logic. The EnerCmed project provides the first transnational empirical test of this paradigm across five Mediterranean port cities.

The long-term vision articulated through this work is the establishment of green harbors across Mediterranean port cities, where energy transition, urban climate resilience, and social equity are pursued as a coordinated, mutually reinforcing strategic goal rather than as separate sectoral objectives. EnerCmed represents a concrete, measurable first step toward this vision, and the lessons learned from its five pilot actions and sixteen follow-up cities will provide a documented roadmap for future initiatives across the Mediterranean region.

Author Contributions

Conceptualization, J.A.B., D.B., and C.S.; methodology, J.A.B., D.B., and C.S.; formal analysis, J.A.B., J.P., and E.F.; investigation, E.F., S.M., E.P., and J.P.; resources, J.A.B. and C.S.; data curation, S.M., J.P., and D.B.; writing—original draft preparation, E.F., S.M., E.P., and J.P.; writing—review and editing, J.A.B., D.B., E.F., S.M., J.P., and C.S.; visualization, J.A.B., S.M., and E.F.; supervision, J.A.B., D.B., and C.S.; project administration, C.S.; funding acquisition, C.S. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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